

Enteropathy-Associated T-Cell Lymphoma in Celiac Disease: A Comprehensive Epidemiological and Clinical Analysis

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Abstract—Enteropathy associated T-cell lymphoma (EATL) is an uncommon and fast growing malignancy of the gastrointestinal tract, generated from intraepithelial T- cell lymphocytes closely related to celiac disease, particularly type II refractory celiac disease. The disease causes severe difficulty in absorbing nutrients and spreads quickly in the small intestine and it causes life threatening complications such as intestinal rupture and haemorrhage, which often limits treatment options. The latest advancement in the World Health Organisation i.e. WHO 5th edition classification of haematolymphoid tumours clearly distinguish EATL from monomorphic epitheliotropic intestinal T-cell lymphoma, highlighting differences in development, appearance and genetic changes. This paper provides a complete overview of EATL, encompassing epidemiological information regarding its occurrence and survival patterns derived from population based data. Additionally, it studies important molecular processes including problems in interleukin-15 signalling, problems in transforming growth factor beta pathways and initiating mutations in the JAK- STAT signaling pathway. Advanced detection methods such as high throughput T-cell receptor sequencing is discussed for their role in checking clonal diversity. The limitation of conventional anthracycline based chemotherapy are highlighted and treatment approaches are examined, along with the potential of targeted treatments, including CD30 directed agents such as brentuximab agents. In addition, ethical consideration regarding genetic testing and clinical Stude signs are discussed. This paper aims to provide an refined and incorporated understanding of EATL for improved diagnosis and management.

Keywords—Celiac Disease, EATL, Intraepithelial Lymphocytes, Refractory Celiac Disease, JAK-STAT pathway targeted therapy.

I. INTRODUCTION

Extra nodal lymphomas most commonly occurs in the gastrointestinal tract and largely derived from B-cells. However, largely intestinal T-cell lymphomas are uncommon, representing approximately 13-15 % of gastrointestinal lymphomas and less than 1% of all non-Hodgkin lymphomas across the world [2]. Among these, enteropathy associated T-cell lymphomas (EATL) is considered as a fast growing and biologically complex cancer, frequently arising as a result of long-term inflammation associated with celiac disease [1]. It shows how long term inflammation from an autoimmune condition can lead to cancer.

In medical terms, EATL is marked by rapid disease progression, involvement of multiple regions of the small intestine and frequent occurring complications such as intestinal perforation and bleeding. These factors lead to a poor outcome, with a median survival about 7 to 10 months and five year survival rates remaining below 20% [3]. The condition becomes worse due to severe malnutrition and hypoalbuminemia and cachexia, which reduce patient's ability to tolerate high intensity chemotherapy.

Recent advances in molecular biology have advanced our understanding of how EATL develops. In particular, mutations in the JAK-STAT signaling pathway, as well as improved characterization of T cell clonality, have provided the foundation for developing targeted therapeutic approaches [5]. The classification of EATL and its management will be redefined through the integration of epidemiological data and the associated genetic basis of the disease.

II. Epidemiology of Celiac Disease and EATL

Epidemiological aspects of enteropathy associated T-cell lymphoma have direct ties to the global incidence of celiac disease, the genetic risk(s) for developing celiac disease, and the testing rates for diagnosis of celiac disease, and until recently enteropathy associated T-cell lymphoma has been a very rare complication of celiac disease. In general, due to the low incidence and to changing diagnostic criteria, there has not been a lot of population-based epidemiological data available on enteropathy associated T-cell lymphoma. However, recent analyses of the Surveillance, Epidemiology, and the End Results (SEER) database from 2000 to 2021 have provided more reliable information on incidence trends and survival outcomes [9]. The current age standardized incidence of EATL, is estimated to be approximately 0.014 per 100,000 individuals [9]. In spite of its rarity, studies indicate a progressive increase in incidence, with an annual percent change Of approximately 2.63% [9]. This increase may result form enhanced detection of celiac disease, advancements in histopathological and molecular diagnostic methods, which enables more precise classification of intestinal T-cell lymphomas [2].

EATL mostly affects older adults, with a median age of onset typically between the sixth and seventh decades of life, averaging around 57-59 years [1]. Particularly, celiac disease is more common in females, EATL shows a consistent male predominance, a phenomenon that remains insufficiently explained [10]. Regionally, the distribution if EATL represents the prevalence of HLA-DQ2 and HLA-DQ8 haplotypes. It is most commonly found in Europe, where it represents approximately 9.1% of peripheral T-cell lymphomas, followed by North America at 5.8% [2]. In contrast it is rare in uncommon in Asian populations, accounting for only 1.9% cases, where many intestinal T-cell lymphomas are refined as monomorphic intestinal T-cell lymphoma, a distinct entity not related to celiac disease [2].

III. LITERATURE REVIEW

A. Evolution of Clinical and Molecular Understanding

The understanding of Enteropathy associated T-cell lymphoma (EATL) has evolved from early reports describing “malignant histiocytosis of the intestine” to a well defined T-cell malignancy with a significant genetic contribution. Early studies build a clear a link between long standing, often

untreated celiac disease and the development of fast growing and severe intestinal lymphoma [12]. A major improvement came from a retrospective study by Malamut et al., which highlighted important prognostic factor, including the types of underlying celiac disease and the patients nutritional status [6]. The study showed that higher serum albumin levels are related with improved survival, highlighting the impact of malnutrition on treatment outcomes. It has also been suggested that combining surgery and chemotherapy may improve survival in some patients [6].

Additional information were provide by the international peripheral T-cell lymphoma project, while Analysed multiple cases and distinguished between different subtypes of EATL based on clinical and immunological features [2]. The study confirmed poor survival outcomes and demonstrated that disease specific prognostic indices are more accurate than general lymphoma scoring systems [2]. In addition, molecular studies , including T-cell receptor sequencing, have shown that progression from refractory celiac disease to EATL involves uncontrolled clonal expansion of abnormal T-cells [8]. Recent genetic findings from Asian population suggests overlapping features between EATL subtypes, indicating that the disease may exits along a biological spectrum instead of clearly separate entities.

B. SUMMARY AND RESEARCH GAP

Table I

SUMMARY OF LITERATURE AND RESEARCH GAPS

Research Area	Summary	Research Gaps
Epidemiology & Clinical Trials	Rare, aggressive T-cell lymphoma with rising incidence and poor survival.	Lack of clinical trial data makes treatment decisions difficult [12].
Molecular Pathogenesis	Disease progression is linked to IL-15 overexpression, JAK-STAT mutations, and chronic inflammation in celiac disease.	The exact transition from celiac disease to lymphoma is not fully understood and needs further molecular research.
Diagnostic & Surveillance Protocols	Diagnosis depends on biopsy and PET-CT, often delayed until severe complications.	There is no standardized screening strategy or reliable early detection markers for high-risk patients.

Therapeutic Strategies	Current treatment includes chemo-immunotherapy and stem cell transplantation approaches.	The best treatment sequence is still unclear, and more targeted therapies are needed for resistant cases.
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IV. PATHOGENESIS OF EATL

Enteropathy-associated T-cell lymphoma (EATL) arises from a chronic autoimmune process that eventually picks up the genetic instability seen in aggressive lymphoma. The sequence starts with a gluten-triggered immune response in people who carry the HLA-DQ2 or HLA-DQ8 alleles [1]. Peptides containing gluten will not become completely digested, and tissue transglutaminase alters gluten peptides to attach more readily to antigen presenting cells which leads to CD4+ T-helper cells activation, producing sustained chronic inflammation in the intestine [2].

Mainly, the disease progression is primarily mediated by the use of cytotoxic CD8 + intraepithelial lymphocytes (IELs) activated by elevated levels of interleukin-15 (IL-15). IL-15 maintains the survival of these lymphocytes, modifies normal immune regulation and causes damage to the intestinal epithelium including the villi [3]. The shift toward malignancy occurs in type II refractory celiac disease, when abnormal clonal lymphocytes expand and stop depending on gluten stimulation [4]. These cells have an altered immunophenotype and gradually accumulate genetic mutations. Persistent IL-15 signaling also helps them avoid apoptosis by activating anti-apoptotic pathways [5].

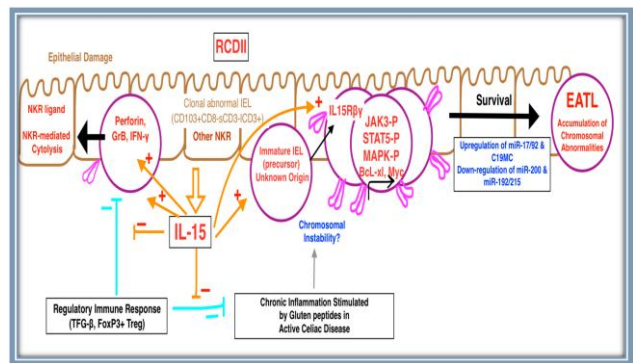


Fig. 1. Pathogenesis pathway of enteropathy-associated T-cell lymphoma (EATL).

V. Molecular and Genetic Mechanisms

The progression from chronic celiac disease to enteropathy-associated T-cell lymphoma (EATL) involves chromosomal abnormalities, genetic mutations, and epigenetic changes. These alterations give EATL a molecular profile that helps distinguish it from other peripheral T-cell lymphomas and may identify targets for treatment [7]. Fig. 2 shows the sequence

from celiac disease to refractory disease and, eventually, EATL.

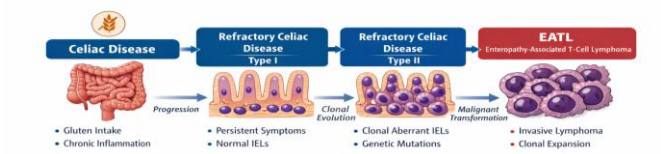


Fig. 2. Progression from celiac disease to refractory celiac disease and EATL.

Cytogenetic studies have reported several recurrent chromosomal abnormalities in EATL. Gain of chromosome 1q is common in type II refractory celiac disease and usually remains present in EATL; this region includes proliferation-related genes such as *CKS1B* [15]. Other changes include gains of 5q and amplifications of 9q, a region that contains immune-regulatory genes such as *CARD9* [10], [11]. Loss of 16q has also been associated with tumor progression [10].

At the molecular level, EATL often shows constitutive activation of the JAK-STAT signaling pathway. Mutations in *JAK1*, *JAK3*, and *STAT3* can keep this pathway active even without cytokine stimulation, which gives tumor cells a growth advantage [7]. Loss of negative regulators such as *SOCS1* can further amplify the signal [7].

Another aspect contributing to the development of EATL is disruption of epigenetics. Ultimately, *TET2* mutations could be used to affect DNA methylation levels due to perturbations caused by mutations of *VAV1* which would affect T-cell signalling [7]. MicroRNAs may also be affected with an increase in the expression of the miR-17/92 cluster and decreases in expression of the miR-200 family which promotes both survival and progression of the tumour cells in EATL [4].

Table II

Genetic Target	Type of Aberration	Frequency	Primary Functional Consequence
1q22-q44	Copy Number Gain	> 70%	Acceleration of cell cycle progression via <i>CKS1B</i>
9q31.3 - qter	Copy Number Gain	60 - 70%	Modulation of innate immune signaling via <i>CARD9</i>

16q12.1	Chromosomal Deletion	~ 25%	Loss of critical tumor suppression mechanisms
STAT3	Somatic Missense Mutation	> 50%	Constitutive cellular proliferation and survival
JAK1	Somatic Missense Mutation	> 40%	Cytokine-independent oncogenic signaling
SOCS1	Somatic Nonsense Mutation	Variable	Total loss of JAK-STAT pathway inhibition

VI. WHO CLASSIFICATION OF INTESTINAL T-CELL LYMPHOMAS

The current WHO classification describes primary intestinal T-cell lymphomas using clinical, microscopic, immune-marker, and molecular findings [16], [23]. EATL and MEITL are now treated as separate diseases, rather than as the older “type I” and “type II” forms of EATL [16], [23].

EATL is an aggressive lymphoma that comes from intraepithelial T cells and is strongly linked to celiac disease [1], [5]. Under the microscope, it usually has mixed, irregular-looking tumor cells with many inflammatory cells around them [10]. The tumor cells are commonly CD3-positive and CD7-positive, usually lack CD5, and may or may not express CD8 [5], [10]. They often show cytotoxic markers such as TIA-1 and granzyme B, and CD30 expression is also common [10]. Because of this, EATL can sometimes be confused with other lymphoid cancers, so diagnosis needs both morphology and immunophenotyping [10], [16].

MEITL was earlier called type II EATL, but now it is taken as a separate intestinal T-cell lymphoma. It is usually not connected with celiac disease and is seen more commonly in Asian populations [11], [16], [24]. Compared to EATL, MEITL has more same-looking tumor cells and clear epitheliotropism, which means tumor cells collect inside the intestinal lining [16], [24]. These cells are mainly CD8-positive and CD56-positive [11], [24]. Genetic studies also identified alterations in JAK-STAT, SETD2 and sometimes MYC, which helps to differentiate MEITL from classic EATL [24].

VII. CLINICAL PRESENTATION AND DIAGNOSTIC APPROACHES

Enteropathy-associated T-cell lymphoma (EATL) is hard to detect early because it can look like celiac disease has suddenly become active again. Many patients are in their late fifties or sixties. They may have long-lasting stomach pain, diarrhea, and major weight loss even when they follow a gluten-free diet [1]. Some patients also have fever, night sweats, and continued weight loss [10].

Diagnosis is often late because these tumors usually develop deep in the small intestine, mainly in the jejunum or ileum. In some patients, EATL is found only after a serious problem such as bowel perforation, bleeding, or blockage, which may need emergency surgery [5]. Although EATL mainly affects the intestine, advanced disease can spread to other parts of the body, including the liver, lungs, bone marrow, or skin [2].

Unusual symptoms can make diagnosis even harder. Rare cases with skin involvement or unusual molecular changes, such as DUSP22 rearrangements, may look like other diseases. In these cases, doctors usually need tissue examination and molecular testing to confirm the diagnosis [3], [18].

Diagnosing EATL needs strong clinical suspicion, imaging, and biopsy. CT scans and 18F-FDG PET-CT can help find intestinal lesions and separate lymphoma from inflammation [1]. Getting a tissue sample may need deep enteroscopy, especially double-balloon enteroscopy. Video capsule endoscopy can also show small-bowel lesions, but it must be used carefully because the capsule can get stuck if the bowel is narrowed [1].

VIII. TREATMENT STRATEGIES

Managing enteropathy-associated T-cell lymphoma (EATL) is complicated because patients are typically malnourished and weak at diagnosis. Chemotherapy, surgery, and nutritional support are all required to effectively manage this disease [5].

An important part of the management of EATL is surgical intervention. It is essential in emergency situations such as intestinal perforation or obstruction and may also improve outcomes when used electively to remove the primary tumor, thereby reducing the risk of complications during chemotherapy [5].

Conventional anthracycline-based regimens such as CHOP have shown limited effectiveness in EATL, with low survival rates and short response duration [5]. Attempts to intensify these regimens have not significantly improved outcomes and are associated with increased toxicity [19].

Improved survival has been observed with intensive chemotherapy protocols followed by autologous stem cell

transplantation. Regimens such as IVE combined with high-dose methotrexate and subsequent transplantation have demonstrated better long-term outcomes in selected patients [20]. However, only a subset of patients can tolerate such aggressive treatment.

Targeted therapy has recently emerged as a promising approach. Brentuximab vedotin, a CD30-directed antibody-drug conjugate, has shown significant efficacy in CD30-positive lymphomas [17]. Clinical trials, including the ECHELON-2 study, have demonstrated improved survival with brentuximab-based regimens compared to standard CHOP therapy, establishing a new treatment standard in eligible patients [17].

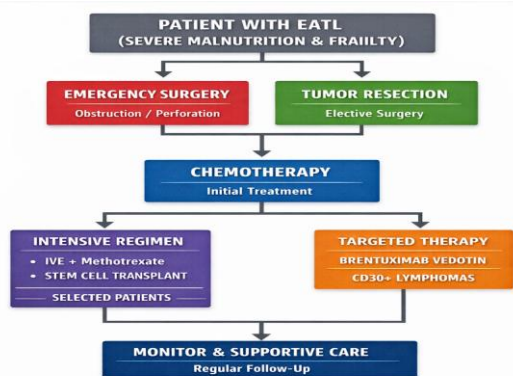


Fig. 4. Treatment flowchart for enteropathy-associated T-cell lymphoma (EATL).

IX. EXPERIMENTAL AND CLINICAL FINDINGS

Advances in molecular analysis and survival modelling have improved the understanding of enteropathy-associated T-cell lymphoma (EATL). High-throughput sequencing of T-cell receptor (TCR) diversity has shown that healthy individuals and patients with controlled celiac disease maintain a diverse, polyclonal T-cell population [8]. In contrast, progression to type II refractory celiac disease and EATL is marked by a significant loss of T-cell diversity, with dominant clonal populations occupying a large proportion of intestinal T-cells [8].

These malignant clones are patient-specific and show no similarity to gluten-reactive T-cells, indicating that tumor growth becomes independent of gluten exposure [8]. This explains why dietary management alone ineffective once malignant transformation occurs.

Survival analysis has also evolved with the use of conditional survival models based on SEER data. While the overall five-year survival rate at diagnosis is low (approximately 16%), survival probability improves significantly for patients who survive the initial years after diagnosis [9]. This trend suggests that early death rate is largely due to complications and treatment challenges, where as patients who respond to initial therapy have a better long term clinical outcome [9].

X. DISCUSSION

EATL is a difficult malignancy to manage because the diagnosis often depends on noticing a change in the clinical pattern early. In patients with known celiac disease, new or worsening symptoms may be blamed on gluten exposure, poor dietary adherence, or infection, especially when the presentation begins with abdominal pain, diarrhea, and weight loss [1], [12], [22]. That can postpone imaging, endoscopic evaluation, and biopsy. By the time EATL is confirmed, some patients already have advanced disease, emergency complications, or poor performance status, all of which can limit treatment options [5], [19], [21].

Molecular studies have also made the classification less straightforward than it once appeared. The WHO classification separates EATL from monomorphic epitheliotropic intestinal T-cell lymphoma (MEITL) but reported immunophenotypic and genetic findings suggest that some cases do not fit neatly into one category [6], [16]. Studies from different populations have described variation in morphology, marker expression, and genomic features, supporting the idea that intestinal T-cell lymphomas may include overlapping biological patterns rather than completely fixed groups [10], [11].

Treatment of EATL is still problematic. Autologous stem cell transplantation and more aggressive chemotherapy and immunotherapy regimens result in better overall survival for some patients than for others. Targeted therapies have emerged as a new treatment option; they target CD30-positive peripheral T-cell lymphomas and have provided some patients with a meaningful response (e.g. brentuximab vedotin) in select patient groups. EATL is influenced by survival signals from inflammatory and immune environments including IL-15 which activates JAK-STAT signaling and allows for the uncontrolled growth of T-cells. Thus, EATL should be viewed as a biologically active, developing process rather than simply a late complication of celiac disease. This perspective could provide future therapies targeting both the neoplasm and the mechanisms that support its growth.

XI. FUTURE SCOPE

A future goal in improving EATL outcomes is to turn molecular findings into more targeted treatments. Currently, over expression and mutated pathways of interleukin-15 have been linked with EATL, and clinical research targets these pathways for therapy development. Recently, early studies have evaluated the efficacy of JAK inhibitors and IL-15 blocking agents to control tumour growth and have yielded promising results. Another possible pathway to prevent EATL involves the nature of the malignant lymphoma. Specifically, targeting specific variant T-cell receptors exhibited by malignant lymphoma may lead to a selective elimination of the EATL tumour cells while maintaining normal T-cell functionality.

Also, new techniques such as liquid biopsy will provide new avenues for EATL early detection and follow-up assessment.. Detection of circulation tumour DNA and specific T-cell receptors patterns could help identify disease development at early stage and improve post treatment tracking. Integration of these molecular tools with advanced data analysis methods may further increases diagnostic accuracy and reduce delays in clinical management.

XII. CONCLUSION

EATL, or enteropathy-associated T-cell lymphoma, is a rare and fast growing cancer, linked with celiac disease. Outcomes are often poor, because it progress fast, cause serious complications, and does not respond well with standard chemotherapy. Recent updates in World Health Organization (WHO) classification has helped to separate EATL from similar intestinal T-cell lymphomas. These updates also give more attention to molecular changes in the disease, including IL-15 activity and changes in JAK-STAT pathway.

Despite newer approaches, including intensive therapy and targeted agent like brentuximab vedodin, have improved outcomes in selected patients, early detection remains the most important factor for survival. Future initiative should focuses on the development of consistent and valid biomarkers, enhances diagnostic approaches and further developing targeted therapies to enhance disease control.

XIII. ACKNOWLEDGEMENT

The authors would like to express their sincere gratitude to their project supervisor for her continuous support and guidance throughout this research. Her valuable suggestions and constructive feedback at every stage greatly helped improve the quality and direction of the study.

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